

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			large surface area/ large area for absorption	1			1		
	(b)			Higher concentration of water outside the cell /lower concentration of solutes outside (1) Water moves into the cell (down a concentration gradient)(1) Through a selectively permeable membrane (1) via osmosis (1)	4			4		
	(c)			Indicative content: • lack of oxygen therefore less aerobic respiration • less energy available • less ATP produced; • active transport of mineral salts • transporting from low to high concentration • Requires ATP/ energy • there is mineral/ nutrient deficiency • poor growth - lack of nitrates • yellow leaves - lack of potassium 5-6 marks All mineral deficiencies should be accounted for as well as linking oxygen lack of oxygen with less ATP from active transport needed for 6 marks. Use of energy rather than ATP gains 5 marks <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	2	4		6		